Problem 1

1. At the isoelectric point the net charge is zero, hence in the DLVO formula there is no electrostatic term, leading to a single-minimum potential corresponding to immediate aggregation. Quantitatively, the Gouy-Chapman length is $l_{GC} = 1/(2\pi l_B \sigma) \to \infty$ (since $\sigma = 0$), from which the electrostatics parameter is $\gamma = l_{GC}/l_D - \sqrt{(l_{GC}/l_D)^2 + 1} \to 0$ and $Z = 16 k_B T \gamma^2 / l_B \to 0$. Hence, the condition for the appearance of a secondary minimum, $Ae^2/(24 l_D Z) < 1$, is not satisfied.

2. Electrostatic repulsion counterbalances the van der Waals attraction. Stronger electrostatics is expected to disfavour aggregation, promoting the appearance of a secondary minimum in the DLVO potential.

3. The condition for the appearance of a secondary minimum is that $Ae^2/(24 l_D Z) < 1$. With the given ionic strength and temperature, the Debye length is $l_D = 1/\sqrt{8\pi l_B N_A I} \simeq 0.79$ nm. In order to compute Z, we first calculate the Gouy-Chapman length $l_{GC} = 1/(2\pi l_B \sigma) \simeq 0.23$ nm. Then, we compute the parameter $\gamma = l_{GC}/l_D - \sqrt{(l_{GC}/l_D)^2 + 1} \simeq -0.75$. From this, we obtain $Z = 16 k_B T \gamma^2 / l_B \simeq 5.33 \cdot 10^{-11}$ N. Substituting in the condition for the appearance of a secondary minimum, we obtain $Ae^2/(24 l_D Z) \simeq 0.73 < 1$. Since the condition is satisfied, a secondary minimum is present in the DLVO formula, from which we expect the presence of a metastable state competing with the aggregation of particles, i.e. the system will not be subjected to fast aggregation.

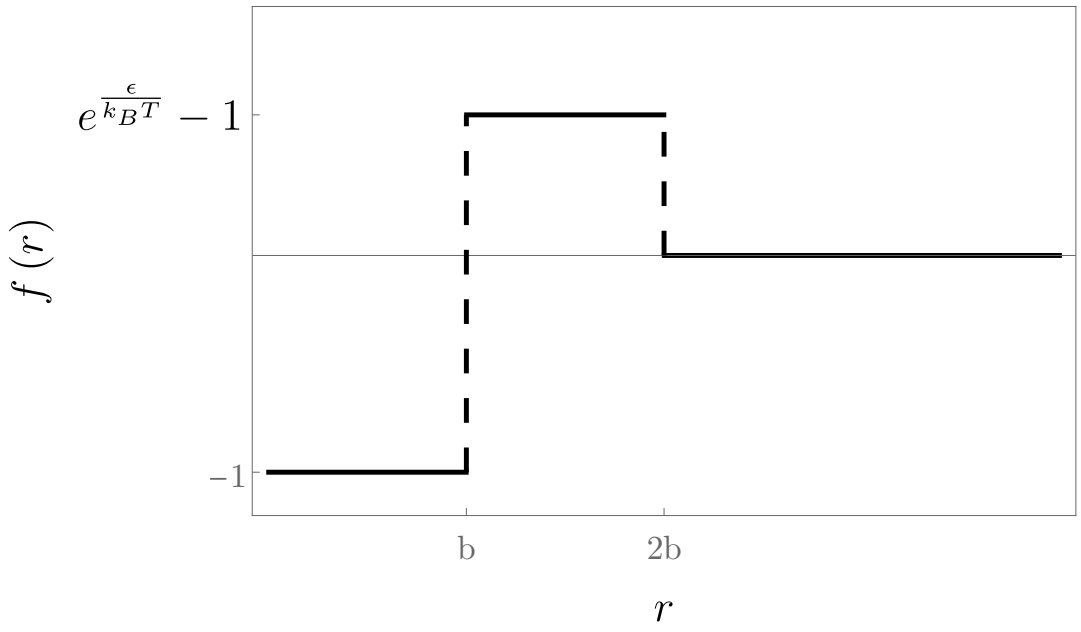
4. Fixing the ionic strength sets the value of l_D. The condition for the appearance of a secondary minimum depends on the charge density by means of Z as $Z > Ae^2/(24 l_D)$. At physiological conditions, the Debye length is $l_D = 1/\sqrt{8\pi l_B N_A I} \simeq 0.79$ nm, hence yielding $Z > 3.88 \cdot 10^{-11}$ N. This gives that $\gamma = -\sqrt{Z l_B/(16 k_B T)} < -0.64$. In turn, this condition imposes for the Gouy-Chapman length $l_{GC} = -l_D(1-\gamma^2)/(2\gamma) < 0.36$ nm. Hence, from the definition of l_{GC} we find $\sigma = 1/(2\pi l_B l_{GC}) > 6.3 \cdot 10^{17}/\mathrm{m}^2$. The total surface of the virus is $4\pi R^2 \simeq 1.13 \cdot 10^{-14}$ m^2. Therefore, for the appearance of a secondary minimum one needs at least 7119 charges, i.e. around 120 charges per unit.

Problem 2

1. The Mayer f-function is defined as $f(r) = e^{-U(r)/k_BT} - 1$, thus

$$f(r) = \begin{cases} -1 & \text{if } r < b \\ e^{\frac{\epsilon}{k_BT}} - 1 & \text{if } b \leq r < 2b \\ 0 & \text{if } r \geq 2b \end{cases}$$

The Mayer function is plotted in Fig.1.



2. ϵ physically represents the depth of the potential well, hence larger values of this parameter result in a stronger attractive part of the potential. Consequently, a larger temperature range will correspond to poor-solvent conditions, i.e. the θ-temperature is shifted to larger values.

3. The excluded volume is obtained by integrating the Mayer f-function over the space, i.e. $v = -4\pi \int_0^\infty r^2 f(r) dr$:

$$v = 4\pi \int_0^b r^2 dr - 4\pi \left(e^{\epsilon/k_BT} - 1\right) \int_b^{2b} r^2 dr = \frac{4}{3}\pi b^3 \left[1 - 7\left(e^{\epsilon/k_BT} - 1\right)\right] = \frac{4}{3}\pi b^3 \left(8 - 7e^{\epsilon/k_BT}\right) .$$

4. Based on the previous formula, the θ-temperature of the system is obtained by definition as the temperature for which $v = 0$:

$$\frac{4}{3}\pi b^3 \left(8 - 7e^{\epsilon/k_B\theta}\right) = 0 \Rightarrow \theta = \frac{\epsilon}{k_B \ln(8/7)} .$$

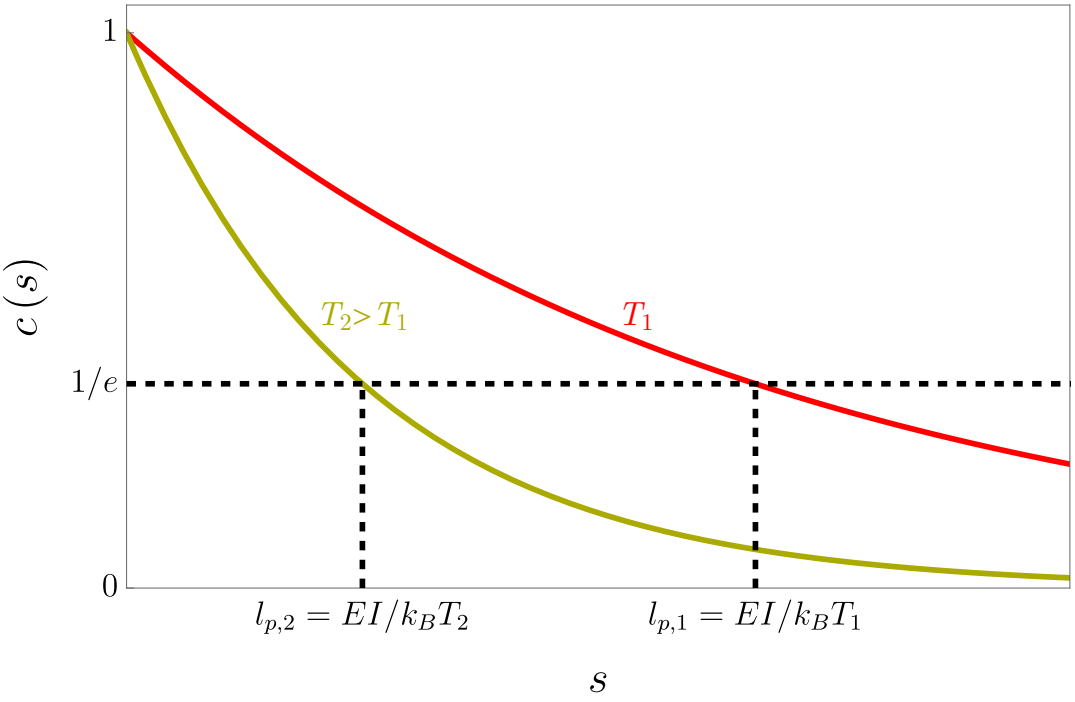
For $T > \theta$, one has $v > 0$, corresponding to good-solvent conditions. Analogously, for $T < \theta$ we find poor solvent conditions corresponding to $v < 0$.

Problem 3

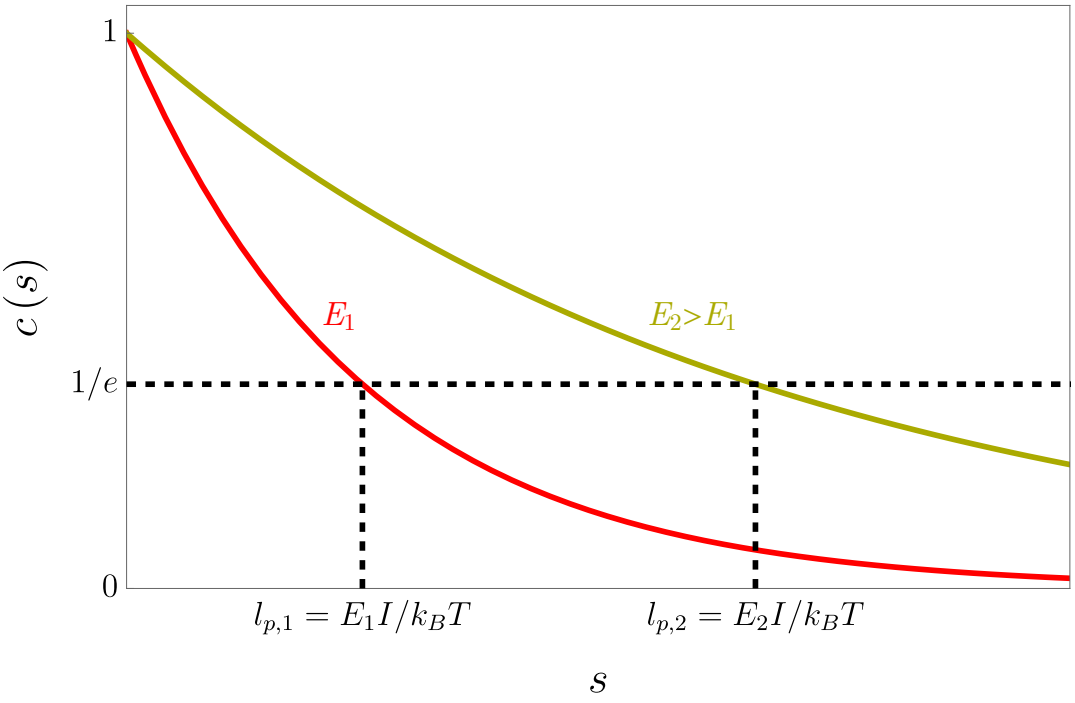
1. By definition of persistence length, the tangent-vector correlation function $c(s)$ is given by $c(s) = \exp(-s/l_p)$, where s is the contour length. Since $l_p = EI/k_BT$ is the scale setting the fall of the exponential decrease, we see that $c(s)$ decreases more fastly for larger temperature. This is in line with the qualitative interpretation of l_p as setting the typical length of thermally-induced bending: larger temperature results in lower l_p, thus more enhanced thermally-induced bending. Quantitatively, substituting $l_p = EI/k_BT$ we have

$$c(s) = e^{-k_BT\frac{s}{EI}}\,, \tag{1}$$

which highlights the starker decrease of the correlation function with higher temperature. The figure below shows the qualitative behavior of the correlation function with two temperatures $T_2 > T_1$. From the formula above, in the limit $T \to 0$ one obtains $c(s) = 1$ independently of s. This is expected, as at zero temperature there is no entropy contribution to the free energy, thus the wormlike chain simply minimizes its bending energy by configuring as a straight rod.



2. A similar reasoning as in the previous point applies. In this case, the Young's modulus appears in the numerator of the persistence length, i.e. a stiffer material results in a larger l_p and a slower decay of the correlation function. One obtains a plot as in the figure below. The effect of a larger radius is qualitatively similar to that of a larger Young's modulus.



3. It is more convenient to improve the precision on the radius, as it appears with a power of 4 in the formula, while the Young's modulus affects the persistence length only linearly.

4. The Young's modulus can be obtained by inverting the formula for the persistence length:

$$E = \frac{k_B T l_p}{I} = \frac{4 k_B T l_p}{\pi r^4} = \frac{4 \cdot 1.38 \cdot 10^{-23} \frac{\text{J}}{\text{K}} \cdot 300 \text{ K} \cdot 50 \cdot 10^{-9} \text{ m}}{\pi \cdot (10^{-9} \text{ m})^4} \simeq 0.26 \text{ GPa}\,, \qquad (2)$$

where we assumed $T = 300$K. This places DNA as a material with stiffness between that of rubber and nylon.

Problem 4

1. The typical time in one-dimensional diffusion can be estimated by considering the time needed for the mean-squared displacement to be equal to L, i.e. $L^2 = 2D\tau \Rightarrow \tau = L^2/(2D)$. The numerical prefactor has little importance in this context, thus the scaling argument $\tau \sim L^2/D$ is also an acceptable solution.

2. The typical time associated with these values is

$$\tau = \frac{(0.1\ \mathrm{m})^2}{2 \cdot 10^{-9}\ \frac{\mathrm{m}^2}{\mathrm{s}}} = 5 \cdot 10^6\ \mathrm{s}\,. \tag{3}$$

   This timescale corresponds to around two months and is clearly incompatible with our everyday experience, indicating that a different transport mechanism is responsible for the spreading of the colour additive.

3. The average squared radius of gyration of a freely-jointed chain of size N and step size b is obtained as $\langle R_g^2 \rangle = Nb^2/6$. To simplify notation, we denote $R_g = \sqrt{\langle R_g^2 \rangle}$, hence the previous formula reads $R_g^2 = Nb^2/6$. The typical time to travel this distance via diffusion is computed as in the previous points as $\tau = R_g^2/(6D)$, where the factor of 6 accounts for the three-dimensional nature of the system. Within the Rouse model, the diffusion coefficient of the polymer is obtained via the Einstein's relation as $D = k_B T/\gamma = k_B T/(N\gamma_0)$. Joining these results together we thus obtain

$$\tau = \frac{R_g^2}{6D} = \frac{\frac{Nb^2}{6}}{6 \cdot \frac{k_B T}{N\gamma_0}} = \frac{b^2\gamma_0}{36 k_B T} \cdot N^2 \propto N^2 \tag{4}$$

4. Compared to the previous point, within the Zimm model the diffusion coefficient is obtained as

$$D = \frac{k_B T}{6\pi\eta R_g}\,, \tag{5}$$

   where η is the viscosity of the solvent. Proceeding as in the case of the Rouse model, here we find

$$\tau = \frac{R_g^2}{6D} = \frac{R_g^2}{6 \cdot \frac{k_B T}{6\pi\eta R_g}} = \frac{\pi\eta}{k_B T} \cdot R_g^3 = \frac{\pi\eta}{k_B T} \cdot \left(\frac{Nb^2}{6}\right)^{\frac{3}{2}} \propto N^{\frac{3}{2}}\,. \tag{6}$$

   In terms of scaling, we see that τ increases now more slowly than in the Rouse model, suggesting that the presence of hydrodynamic interactions is helping the diffusion of the polymer.

IMPORTANT NOTE: do not take this solution as a rigorous derivation of the Rouse and Zimm models.

Problem 5

1. In the case of a massless particle, the gravity has no effect at all. We thus expect the height to be uniformly distributed in $[0, h]$. For a very heavy particle, and since its density is larger than that of water, we expect that the particle will sink at the bottom of the box.

2. The probability distribution is proportional to the Boltzmann's weight of the potential energy, $p(z) = e^{-\beta\widetilde{m}gz}/\mathcal{Z}$, where $\mathcal{Z}$ is the configurational partition function. We find easily that

$$\mathcal{Z} = \int_0^h e^{-\beta\widetilde{m}gz} dz = \frac{1 - e^{-\beta\widetilde{m}gh}}{\beta\widetilde{m}g}\,, \tag{7}$$

hence

$$p(z) = \frac{\beta\widetilde{m}g}{1 - e^{-\beta\widetilde{m}gh}} e^{-\beta\widetilde{m}gz}\,, \tag{8}$$

which is the formula in the text of the problem. In the case of a massless particle, both the numerator and the denominator of the prefactor go to zero. Taylor expansion of the denominator gives $1 - e^{-\beta\widetilde{m}gh} \simeq \beta\widetilde{m}gh$, thus the prefactor reduces to $1/h$ as $\beta\widetilde{m}gh \to 0$. Therefore, we get $p(z) \simeq 1/h$, which is a uniform distribution in the range $[0, h]$, as expected.

3. The average height is found as

$$\langle z \rangle = \int_0^h z\, p(z) = \frac{\beta\widetilde{m}g}{1 - e^{-\beta\widetilde{m}gh}} \int_0^h z\, e^{-\beta\widetilde{m}gz} dz\,. \tag{9}$$

Employing the formula suggested in the text of the problem, we can thus write:

$$\langle z \rangle = \frac{\beta\widetilde{m}g}{1 - e^{-\beta\widetilde{m}gh}} \cdot \frac{1 - e^{-\beta\widetilde{m}gh}\,(1 + \beta\widetilde{m}gh)}{(\beta\widetilde{m}g)^2}\,, \tag{10}$$

that is

$$\langle z \rangle = \frac{1}{\beta\widetilde{m}g} \cdot \frac{1 - e^{-\beta\widetilde{m}gh} - e^{-\beta\widetilde{m}gh}\beta\widetilde{m}gh}{1 - e^{-\beta\widetilde{m}gh}}\,. \tag{11}$$

Further rearranging gives

$$\langle z \rangle = \frac{1}{\beta\widetilde{m}g} - \frac{h}{e^{\beta\widetilde{m}gh} - 1} = h \cdot \left(\frac{1}{\beta\widetilde{m}gh} - \frac{1}{e^{\beta\widetilde{m}gh} - 1} \right)\,, \tag{12}$$

which is the formula in the text of the problem. From the formula, it is evident that $\langle z \rangle / h$ depends only on the dimensionless combination $\beta\widetilde{m}gh$. In the case of infinite mass or, more precisely, when $\beta\widetilde{m}gh \to \infty$, we get easily that $\langle z \rangle \to 0$, i.e. that the particle

sinks as expected qualitatively. In the opposite case of $\beta\widetilde{m}gh \to 0$, the formula is a bit trickier. It is best to rewrite Eq. 12 as

$$\frac{\langle z\rangle}{h} = \frac{e^{\beta\widetilde{m}gh} - 1 - \beta\widetilde{m}gh}{\beta\widetilde{m}gh \cdot (e^{\beta\widetilde{m}gh} - 1)}\,. \tag{13}$$

Then, we consider the Taylor series around $\beta\widetilde{m}gh = 0$ and retain the leading terms:

$$\frac{\langle z\rangle}{h} \simeq \frac{1 + \beta\widetilde{m}gh + \frac{1}{2}\left(\beta\widetilde{m}gh\right)^2 - 1 - \beta\widetilde{m}gh}{\beta\widetilde{m}gh \cdot (1 + \beta\widetilde{m}gh - 1)} = \frac{1}{2}\,, \tag{14}$$

thus yielding $\langle z\rangle = h/2$. This is the expected outcome for a uniform distribution of height in the range $[0, h]$, which was the physical expectation in this limit.

4. Assuming the protein to be a sphere of radius $R = 3$ nm and density $\rho = 1350$ kg/m^3, its mass is obtained as

$$m_{\text{protein}} = \frac{4}{3}\pi R^3 \cdot \rho = \frac{4}{3}\pi\left(3 \cdot 10^{-9}\text{m}\right)^3 \cdot 1350\frac{\text{kg}}{\text{m}^3} \simeq 1.53 \cdot 10^{-22}\ \text{kg} \simeq 92\ \text{kDa}\,, \tag{15}$$

where the latter conversion was made remembering that 1 Da = 1 g/mol. Accounting for buoyancy, we obtain as effective mass $\widetilde{m} = m \cdot (1 - 1000/1350) \simeq 3.96 \cdot 10^{-23}$ kg. The dimensionless combination thus yields

$$\beta\widetilde{m}gh \simeq \frac{1}{1.38 \cdot 10^{-23}\,\frac{\text{J}}{\text{K}} \cdot 300\ \text{K}} \cdot 3.96 \cdot 10^{-23}\ \text{kg} \cdot 9.8\,\frac{\text{m}}{\text{s}^2} \cdot 50 \cdot 10^{-6}\ \text{m} \simeq 4.7 \cdot 10^{-6} \ll 1\,, \tag{16}$$

which implies that the mass of the protein is too small and the effect of gravity on adsorption is negligible. This can be seen explicitly substituting this value into Eq. 12, which gives

$$\frac{\langle z\rangle}{h} \simeq 0.5 \Rightarrow \langle z\rangle \simeq 25\mu\text{m} \gg 3\text{nm}\,. \tag{17}$$

A similar calculation for a bacterium yields a mass equal to

$$m_{\text{bacterium}} = \frac{4}{3}\pi\left(1 \cdot 10^{-6}\text{m}\right)^3 \cdot 1100\frac{\text{kg}}{\text{m}^3} \simeq 4.6 \cdot 10^{-15}\ \text{kg}\,, \tag{18}$$

setting the mass scale of a bacterium on the order of the picograms. The effective mass is $\widetilde{m} = m \cdot (1 - 1000/1100) \simeq 4.2 \cdot 10^{-16}$ kg, which results in a dimensionless combination equal to

$$\beta\widetilde{m}gh \simeq \frac{1}{1.38 \cdot 10^{-23}\,\frac{\text{J}}{\text{K}} \cdot 300\ \text{K}} \cdot 4.2 \cdot 10^{-16}\ \text{kg} \cdot 9.8\,\frac{\text{m}}{\text{s}^2} \cdot 50 \cdot 10^{-6}\ \text{m} \simeq 49.6 \gg 1\,. \tag{19}$$

In this case, we thus expect the bacterium to sink (compared to the length scale h). To check if this is sufficient to have a gravity-driven adsorption, we substitute into Eq. 12 obtaining

$$\frac{\langle z\rangle}{h} \simeq 0.02 \Rightarrow \langle z\rangle \simeq 1\mu\mathrm{m}\,, \tag{20}$$

which is indeed comparable to the size of the bacterium. We note that this is a numerical coincidence: in the limit $\beta\widetilde{m}gh \gg 1$, the leading term in Eq. 12 is $\langle z\rangle \simeq 1/(\beta\widetilde{m}g)$. A particle of similar density but e.g. 5 times smaller in mass would yield $\beta\widetilde{m}gh \simeq 10$, which is still large compared to 1. Yet, the average height would now be 5 times larger, i.e. around 5 μm. The radius would be instead 1 $\mu\mathrm{m}/5^{1/3} \simeq 0.6\,\mu$m, hence the adsorption would only be weakly affected by gravity.

Problem 6

1. The net repulsion between the like charges along the chain are expected to induce a stiffening of the polymer

2. The Debye length is equal to
$$l_D = \frac{1}{\sqrt{8\pi N_A l_B I}} \,. \tag{21}$$
Remembering that 1 mM = 1 mol/m^3, for $I = 150$ mM we get
$$l_D = \frac{1}{\sqrt{8\pi \cdot 6.022 \cdot 10^{23}\ \mathrm{mol}^{-1} \cdot 7 \cdot 10^{-10}\ \mathrm{m} \cdot 150\ \mathrm{mol/m}^3}} \simeq 0.8\ \mathrm{nm}\,. \tag{22}$$
Substituting in the formulae, this gives $l_{p,e} \simeq 0.22$ nm for the OSF theory, while $l_{p,e} \simeq 1.6$ nm for the BJ theory. In this case, we thus see that the BJ theory predicts a stronger effect of electrostatics. A similar calculation at 1 mM yields $l_D \simeq 9.7$ nm, which gives $l_{p,e} \simeq 34$ nm for the OSF theory, while $l_{p,e} \simeq 19$ nm for the BJ theory. We see that the situation is now reversed, with the OSF theory predicting a stronger electrostatic component of the persistence length.

3. For the OSF theory to predict a stronger effect than the BJ theory, the following condition needs to be fulfilled:
$$\frac{l_D^2}{4l_B} > 2l_D \Rightarrow \frac{l_D}{8l_B} > 1\,. \tag{23}$$
Substitution of the formula for l_D yields:
$$\frac{l_D}{8l_B} = \frac{1}{8l_B\sqrt{8\pi N_A l_B I}} > 1 \Rightarrow 512\pi N_A l_B^3 I < 1 \Rightarrow I < I_c \equiv \frac{1}{512\pi N_A l_B^3}\,, \tag{24}$$
which is the formula reported in the text of the problem. Substituting the numerical values, one obtains $I_c \simeq 3$ mM.

4. According to the OSF theory, and substituting the formula for l_D, the overall persistence length is
$$l_p = l_{p,0} + \frac{1}{32\pi N_A l_B^2 I}\,, \tag{25}$$
which is of the form $A/I + B$, thus compatible with the observed behavior. In contrast, the BJ theory would be compatible with a formula like $A/\sqrt{I} + B$. We thus conclude that the qualitative behavior of the data is compatible with the OSF prediction. Quantitatively, one can immediately associate the bare persistence length to B, i.e. $l_{p,0} = B = (25 \pm 2)$

nm. In contrast, it is expected that $A = 1/(32\pi N_A l_B^2)$. We check numerically if this is correct:

$$\frac{1}{32\pi N_A l_B^2} = \frac{1}{32\pi \cdot 6.022 \cdot 10^{23}\ \mathrm{mol}^{-1} \cdot (7 \cdot 10^{-10}\ \mathrm{m})^2} \simeq 3.4 \cdot 10^{-8}\, \frac{\mathrm{mol}}{\mathrm{m}^2}\,. \tag{26}$$

Note that 1 mM = 1 mol/m^3, hence the previous numerical results can be written as $3.4 \cdot 10^{-8}$ mM$\cdot$ m= 34 mM $\cdot$ nm. The value of A is thus compatible with the expectation from the OSF theory.

1. For the Freely-Jointed Chain, we know that $\langle R_e^2 \rangle = Nb^2$ independently of the dimensionality of the system.

2. When changing the dimensionality of the system, only the enthalpic part of the Flory free energy needs to be modified. If the total available area is $A \sim R^2$, then the probability of overlap between any two monomers is $a/A \simeq v/(R^2 \cdot b)$, where $a \sim v/b$ is the "excluded" area. The total number of monomer couples scales as N^2, hence the enthalpic contribution to the free energy is $k_B T \cdot N^2 v/(R^2 \cdot b)$. The entropic part is instead $k_B T R^2/(Nb^2)$ independently of the dimensionality. The Flory free energy thus reads:

$$F \sim k_B T \frac{N^2 v}{R^2 \cdot b} + k_B T \frac{R^2}{Nb^2} \,. \tag{27}$$

   Minimizing with respect to R and dropping numerical prefactors, we find

$$k_B T \frac{N^2 v}{R^3 \cdot b} \sim k_B T \frac{R}{Nb^2} \Rightarrow R \sim v^{\frac{1}{4}} b^{\frac{1}{4}} N^{\frac{3}{4}} \,, \tag{28}$$

   which is the formula reported in the text.

3. Since $3/4 > 3/5$, R increases more rapidly with N in two dimensions than in three. Similarly, $1/4 > 1/5$, so the same applies for the scaling with v. Instead, the impact of the step size b is the opposite: $1/4 < 2/5$, hence R increases more rapidly with b in three dimensions.

4. Substituting $v \sim b_K^2 \cdot r$ gives $R_{2D} \sim r^{1/4} b_K^{3/4} N_K^{3/4}$ in two dimensions, while $R_{3D} \sim r^{1/5} b_K^{4/5} N_K^{3/5}$ in three dimensions. Excluding prefactors, the condition $R_{2D} > R_{3D}$ implies

$$r^{\frac{1}{4}} b_K^{\frac{3}{4}} N_K^{\frac{3}{4}} > r^{\frac{1}{5}} b_K^{\frac{4}{5}} N_K^{\frac{3}{5}} \Rightarrow N_K^{\frac{3}{20}} > b_K^{\frac{1}{20}} r^{-\frac{1}{20}} \Rightarrow N_K > \left(\frac{b_K}{r} \right)^{\frac{1}{3}} \equiv N_c \,, \tag{29}$$

   that is the formula reported in the text. In the case of DNA, the Kuhn length in three dimensions is $2l_p = 100$ nm, so that $N_c \simeq 4.6$. This implies that a contour length of roughly 500 nm is needed before the polymer is larger when confined onto a plane, although the exact value will depend on the prefactors that in this calculation have been neglected.

Problem 8

1. This question has been answered in the theory: if electrostatics is strong enough (large Z and/or small A), a secondary minimum appears and there is a certain transient time in which the system is metastable as a dispersed solution. If electrostatics is weak, only the primary minimum is present and immediate aggregation will take place.

2. We know that
$$Z = \frac{16 k_B T \gamma^2}{l_B}, \tag{30}$$
where $k_B T = 4.14$ pN·nm and $l_B = 0.7$ nm for $T = 300$ K, while $\gamma = l_{GC}/l_D - \sqrt{(l_{GC}/l_D)^2 + 1}$. The surface charge density for this colloid is $\sigma = 0.1$ nm^{-2}, hence the Gouy-Chapman length is
$$l_{GC} = \frac{1}{2\pi l_B \sigma} \simeq 2.27 \text{ nm}. \tag{31}$$
The Debye length is also easily calculated starting from $I = 30$ mM (be careful to use International System units for this calculation to avoid mistakes).
$$l_D = \frac{1}{\sqrt{8\pi l_B N_A I}} \simeq 1.77 \text{ nm}. \tag{32}$$
Combining these two values, we obtain that $\gamma \simeq -0.34$, which gives $Z \simeq 11.2$ pN.

3. The condition for the appearance of a secondary minimum is
$$\frac{A\, e^2}{24 l_D Z} \leq 1. \tag{33}$$
Substituting $A = 0.4 \cdot 10^{-19}$ for the Hamaker constant and the other values computed in the previus point, we obtain for the left-hand size $A\, e^2/24 l_D Z \simeq 0.62$, hence the secondary minimum appears. The free energy is shown in the figure. Changing the radius R does not affect the overall shape of the function, since R appears as a common factor in both the van der Waals and electrostatic terms.

4. Within an Arrhenius approximation, the typical time τ needed to observe aggregation is $\tau = \tau_0 \exp(\Delta F/k_B T)$, where $\Delta F \simeq 2.7 k_B T$ is the free-energy barrier and τ_0 is a prefactor related to the diffusion of the colloid. The latter is estimated here by considering $d_C^2 = 6 D \tau_0$, where D is the diffusion coefficient of the colloid. Within a Stokes-Einstein framework, we have $D = k_B T/(6\pi\eta R)$, hence:
$$\tau_0 = \frac{d_C^2}{6D} = \frac{d_C^2}{6} \cdot \frac{6\pi\eta R}{k_B T} = \frac{\pi \eta\, d_C^2\, R}{k_B T} \simeq 1.3 \cdot 10^{-4} \text{ s}. \tag{34}$$

The typical time needed to observe aggregation is thus $\tau \simeq 1.3 \cdot 10^{-4}$ s $\cdot e^{2.7} \simeq 2 \cdot 10^{-3}$ s. These are just a few milliseconds, so that I will typically observe aggregation in my experiments. If I change the radius, both τ_0 and ΔF need to be rescaled by a factor of 5. For $R = 100$ nm, we thus have $\tau_0 \simeq 6.5 \cdot 10^{-4}$ s and $\Delta F \simeq 13.5 k_B T$, which gives $\tau \simeq 500$ s; this corresponds to roughly 8 minutes, so that I will be able to see aggregation in my experiments. However, if the particle's radius becomes 200 nm, we obtain $\tau_0 \simeq 1.3 \cdot 10^{-3}$ s and $\Delta F \simeq 27 k_B T$, which gives $\tau \simeq 7 \cdot 10^8$ s; this corresponds to about 22 years, so that very few people would be willing to waste their life waiting for aggregation to happen.

